ARTHRITIS & RHEUMATOLOGY
Vol. 69, No. 9, September 2017, pp 1862–1870
DOI 10.1002/art.40178

Characteristics and Management of IgA Vasculitis (Henoch-Schönlein) in Adults

Data From 260 Patients Included in a French Multicenter Retrospective Survey

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Objective. **Data on adult IgA vasculitis (Henoch-Schönlein) (IgAV) are scarce. This survey was designed to better define the clinical spectrum of IgAV and efficacy of treatments in a French patient population.**

Methods. **Data on clinical characteristics, histologic features, and treatment response from 260 patients with IgAV included in a French multicenter retrospective survey were analyzed. Efficacy data were compared using different statistical models.**

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Results. **The mean ± SD age of the patients with IgAV at diagnosis was 50.1 ± 18 years, and 63% of patients were male. Baseline manifestations included purpura (100%), arthralgias/arthritis/myalgia (61%), glomerulonephritis (70%), and/or gastrointestinal involvement (53%). Thirty percent of patients showed renal failure at baseline. In univariate analysis, the response to therapy was 80% (64 of 80) in patients treated with corticosteroids (CS) alone, compared to 77% (23 of 30) in patients treated with CS plus cyclophosphamide (CYC) and 59% (10 of 17) in patients treated with colchicine (*P* = 0.17). Multivariable analysis showed that treatment with CS or CS plus CYC was more effective than colchicine in achieving a response. Efficacy differences were demonstrated using different statistical models: in the multivariable logistic regression model, odds ratio (OR) 3.68, 95% confidence interval (95% CI) 1.10–12.33 (*P* = 0.03); in the inverse probability weighting on propensity score model, OR 3.75, 95% CI 1.28–10.99 (*P* = 0.02). The efficacy of CS plus CYC as compared to CS alone was discordant according to the analytic method used. Analysis with the multivariable logistic regression model did not demonstrate a difference between CS plus CYC and CS alone (OR 0.88, 95% CI 0.29–2.67; *P* = 0.82). In contrast, inverse probability weighting on propensity score showed that CS plus CYC was more effective than CS alone (OR 1.79, 95% CI 1.00–3.20; *P* = 0.049).**

Conclusion. **This series constitutes the largest series of adults with IgAV reported in the literature so far. It provides data on clinical and histologic presentation and therapeutic efficacy, suggesting that CS alone appears to be a reasonable first-line therapy in patients with IgAV, while the benefit of adding CYC to CS remains uncertain.**

IgA vasculitis (IgAV), formerly called Henoch-Schönlein purpura, is an immune complex small vessel vasculitis with IgA1-dominant immune deposits (1). IgAV is the most common systemic vasculitis in childhood, with an annual incidence of 3–26 per 100,000 children (2). In adults, the disease is less common, with an annual incidence of 0.1–1.8 per 100,000 individuals (3–5).

IgAV frequently involves the skin, gastrointestinal tract, joints (arthralgias and/or arthritis), and kidneys (6). Gastrointestinal tract and renal involvement represent the main causes of morbidity and mortality in adults. In a large study of 250 adults with IgAV, 11% of patients reached end-stage renal disease (ESRD), 13% had severe renal failure with an estimated glomerular filtration rate (eGFR) of <30 ml/minute/1.73 m^2, and 14% had moderate renal failure with an eGFR of <50 ml/minute/1.73 m^2 (7). Factors found to be associated with evolution to ESRD include baseline renal function, a baseline proteinuria level of >1 gm/day or 1.5 gm/day, macroscopic hematuria, hypertension, and a proteinuria level of ≥1 gm/day during follow-up (7–9). In a study of a Southern European population, hematuria at disease onset, renal involvement during the course of the disease, anemia at the time of diagnosis, and onset of the disease in summer were more common in patients with renal sequelae (10). Evidence from renal biopsy has shown that the degree of interstitial fibrosis, sclerotic glomeruli, and fibrinoid necrosis are also associated with a poor renal prognosis (7).

Treatment of IgAV has been poorly investigated in adults. Management of IgAV is often targeted to specific symptoms, because the course of the disease is frequently benign and remission can occur spontaneously. However, in the case of severe organ involvement in the presence of organ- and/or life-threatening complications, corticosteroids (CS) and/or immunosuppressive drugs are often initiated. Pillebout et al conducted the only multicenter prospective, open-label trial in adults with IgAV, comparing CS alone to CS plus cyclophosphamide (CYC) for the treatment of patients with severe IgAV (11). Fifty-four patients with biopsy-proven IgAV and severe manifestations, including proliferative glomerulonephritis and/or severe gastrointestinal manifestations, were included. At 12 months, no difference was found between the 2 groups in terms of remission rate, renal outcomes, and adverse events. However, overall survival at 12 months tended to be better in the CS plus CYC group compared to the CS alone group (96% survival rate, 95% confidence interval [95% CI] 89–100% versus 79% survival rate, 95% CI 64–93%; *P* = 0.08). Thus, the optimal therapeutic strategy has yet to be defined in adults.

To better define the clinical spectrum and the efficacy of treatments in IgAV in adults, a nationwide survey was initiated in France in 2013. Data from the 260 cases of IgAV in adults included in this French

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Submitted for publication June 12, 2016; accepted in revised form June 8, 2017.

retrospective survey (known as the IGAVAS) are reported herein.

PATIENTS AND METHODS

Patients. The multicenter retrospective IGAVAS survey was conducted in French university hospitals and general hospitals within the Departments of Internal Medicine, Nephrology, Dermatology, and Rheumatology. The study was performed in accordance with the ethics standards of the Declaration of Helsinki, and was approved by the Institutional Review Boards of each hospital. The inclusion criteria for the study were as follows: age >18 years, diagnosis of IgAV, and diagnosis of IgAV made between January 1990 and January 2015. Patients were considered to have IgAV if they presented with purpura, histologically proven small vessel vasculitis, histologically proven IgA deposits, and involvement of at least 1 organ among the kidneys, joints, or intestinal tract. One exclusion criterion was the presence of IgAV associated with a diagnosis of cancer in the 5 years prior to onset of vasculitis.

Clinical and biologic data. Data on clinical and biologic characteristics were recorded using a standardized form for each patient at the time of the initial evaluation, during follow-up (6 and 12 months after the initial evaluation), and at the end of follow-up. The practitioners in charge of the patients administered the questionnaire. Laboratory assessments included, in particular, determination of serum creatinine levels, urinalysis to screen for hematuria, and a 24-hour urine protein examination. Renal failure was defined as an eGFR of <60 ml/minute/1.73 m^2, assessed with the Modified Diet in Renal Disease equation (12). Proteinuria was defined as a 24-hour urinary protein excretion rate of >0.5 gm/day, and hematuria was defined as >10 red cells/mm^3 in the urine, which was considered to be macroscopic if there were >1,500 red cells/mm^3. Elevated IgA levels were defined as an IgA level of >3.5 gm/liter.

Histologic data. Data on histologic features (findings from skin and renal biopsies) were recorded at the time of diagnosis. Pathology reports on the renal biopsy findings were examined by 2 independent nephrologists (ET and EP) who were blinded with regard to the clinical features. All renal biopsy samples were classified for severity of glomerulonephritis (classes I–V) according to the classification system previously described by Pillebout et al (7), based on the presence of a focal or diffuse distribution of involved glomeruli, extracapillary proliferation, number of glomeruli involved, interstitial fibrosis, proportions of glomeruli involved by crescents, fibrinoid necrosis, and global sclerosis.

Response to therapy. The response to therapy for IgAV was defined by analyzing the course of the following main clinical signs: skin involvement (purpura), articular manifestations (arthralgias and/or arthritis), gastrointestinal symptoms, and renal involvement (normalization or improvement of the eGFR, the levels of proteinuria, and hematuria). Responses were evaluated by 2 independent physicians (AAV and BT) who were blinded with regard to the treatment received.

A complete response was defined as an improvement in all baseline clinical manifestations and, in the case of renal involvement, a decrease in the proteinuria level to <0.5 gm/day, the disappearance of hematuria, and no decrease in the eGFR of more than 20% from baseline. A partial response was defined as an improvement in at least one-half of the baseline clinical manifestations and, in the case of renal involvement, an improvement in the proteinuria level by >50% of the baseline value, either disappearance or no disappearance of hematuria, and no decrease in the eGFR of more than 20% from baseline. All other patients were classified as nonresponders.

Relapse was defined as the reappearance of clinical signs of vasculitis, occurring after a period that was free of symptoms for at least 1 month. Minor relapse was defined as an increase in the dosage of prednisone to no more than 20 mg/day. Major relapse was defined as the addition of an immunosuppressive drug to the regimen or an increase in the dosage of prednisone to more than 20 mg/day. All other patients were considered to have experienced no relapse.

Statistical analysis. Descriptive statistics included the mean ± SD values or median with interquartile range (IQR) as appropriate for continuous variables, and the frequency (percentage) for categorical variables. Univariate analysis included the chi-square or Fisher's exact test as appropriate for comparing categorical variables, and the nonparametric Mann-Whitney test to compare continuous variables. To evaluate the efficacy of the therapeutic regimens, we performed a multivariable logistic regression model to assess factors potentially independently associated with response to therapy. All important factors noted in the literature, as well as those factors associated with being treated or with response to therapy in univariate analyses, were included in the final multivariable logistic regression model. Not all correlated variables were entered in the model, to avoid the risk of collinearity.

To account for a potential indication bias (i.e., patients treated with CS or CYC may have more severe disease than those treated with colchicine only), we also estimated a propensity score using a logistic regression model. The propensity score corresponded to the probability of being treated with CS or CYC, rather than colchicine, based on the patient's characteristics (age, sex, and presence of gastrointestinal bleeding, creatininemia, proteinuria, or necrotic purpura) (13,14). We performed an evaluation of the distribution of propensity scores by treatment group, checking for sizeable overlap among the treatment groups (i.e., it is not possible to use a propensity score when the 2 groups are too different; this is why it is important to check whether the 2 groups overlap).

There are several ways to use the propensity score in statistical analyses, including 1) adjustment on the propensity score as a covariate in the model, 2) matching of individuals treated and those not treated based on their propensity score, and 3) inverse probability weighting on propensity score. The 2 preferred methods are matching and weighting (15). However, it is not always possible to use matching, since participants who are not matched (i.e., patients treated and those not treated for whom the propensity score is not the same) are not included in the analyses. In this study, we used both adjustment on propensity score and inverse probability weighting on propensity score in the logistic regression models to assess whether treatment with CS or CYC was associated with a better response compared to treatment with colchicine, and to assess whether treatment with CYC was associated with a better response compared to treatment with CS alone. Because inverse probability weighting analysis may be sensitive to extreme propensity scores (13,14), we did a sensitivity analysis in which we excluded the 5% of patients with the lowest propensity scores (designated outliers). Adjusted odds ratios (ORs) for all covariates in the model without propensity score are shown in Supplementary Table 1 (available on the *Arthritis & Rheumatology* web site at http://onlinelibrary.wiley.com/doi/10.1002/art.40178/abstract).

RESULTS

Clinical and laboratory characteristics of the patients. Of the 304 patients assessed for eligibility in the IGAVAS survey, 260 patients were included. Forty-four patients were excluded because of absence of proven IgA deposits (n = 27), absence of proven vasculitis (n = 4), concomitant cancer (n = 6), or missing data (n = 7). The main baseline characteristics of the 260 patients with IgAV are shown in Table 1. The mean ± SD age of the patients at diagnosis was 50.1 ± 18 years, and 164 patients (63%) were male. Clinical manifestations included constitutional symptoms in 87 patients (33%), cutaneous involvement with purpura in all patients (100%), arthralgias/arthritis/myalgia in 159 patients (61%), renal involvement in 182 patients (70%), and gastrointestinal involvement in 137 patients (53%). Thirty percent of the patients showed renal failure (eGFR of <60 ml/minute/1.73 m^2) at baseline. Patients with renal involvement displayed a median eGFR of 90 ml/minute/1.73 m^2 (IQR 66–110) and a median proteinuria level of 1.5 gm/day (IQR 0.6–3), and 88% of the patients presented with hematuria. The median serum IgA level was 3.6 gm/liter (IQR 2.7–4.8) and 85 (53%) of 159 patients presented with elevated IgA levels.

Antineutrophil cytoplasmic antibodies were detected in 9 (4%) of 225 patients. Antinuclear antibodies were present in 33 (14%) of 231 patients. No specificity was detected for either of these antibodies (data not shown).

Histologic features. The histologic characteristics of the skin and renal biopsy samples obtained from the patients with IgAV are summarized in Table 2. A skin biopsy was performed in 222 patients (85%), and a renal biopsy was performed in 144 (79%) of the 182 patients with kidney involvement. Evaluation of the skin biopsy specimens demonstrated leukocytoclastic vasculitis in 205 patients (92%). Direct immunofluorescence revealed deposition of IgA and deposition of complement in the blood vessels of the upper dermis in 174 (81%) of 216 patients and 47 (21%) of 222 patients, respectively. Direct immunofluorescence was not performed on 6 skin biopsy specimens. Evaluation of the renal biopsy specimens demonstrated IgA mesangial deposits in 142 (99%) of 144 patients, and extracapillary proliferation in 59 (41%) of 144 patients.

Table 1. Main baseline characteristics of the 260 patients with IgA vasculitis*

Epidemiologic features	
Age at diagnosis, mean ± SD years	50.1 ± 18.6
Male	164 (63)
Clinical manifestations	
Constitutional symptoms	87/260 (33)
Fever	39/260 (15)
Asthenia	54/260 (21)
Weight loss	22/260 (8)
Skin symptoms	260/260 (100)
Purpura	260/260 (100)
Lower limb	260/260 (100)
Upper limb	93/260 (36)
Abdomen	63/260 (24)
Face	8/260 (3)
Necrosis	67/260 (25)
Hemorrhagic blisters	22/260 (8)
Joints	159/260 (61)
Arthralgias	159/160 (100)
Arthritis	26/160 (16)
Myalgias	10/160 (6)
Gastrointestinal tract	137/260 (53)
Abdominal pain	135/137 (99)
Bleeding	43/137 (31)
Diarrhea	36/137 (26)
Nausea	26/137 (19)
Ileus	13/137 (9)
Kidney	182/260 (70)
Edema	49/182 (27)
High blood pressure	40/182 (22)
Macroscopic hematuria	18/182 (10)
Other	13/260 (5)
Peripheral neuropathy	4/260 (2)
Orchiepidymitis	4/260 (2)
Cardiopathy	2/260 (1)
Intraalveolar hemorrhage	1/260 (0.5)
Biologic features	
Serum IgA level, median (IQR) gm/liter	3.6 (2.7–4.8)
Elevated serum IgA level	85/159 (53)
Serum creatinine level, median (IQR) μmoles/liter	77 (65–99)
eGFR, median (IQR) ml/minute/1.73 m^2	90 (66–110)
eGFR <60 ml/minute/1.73 m^2	55/181 (30)
Albumin, median (IQR) gm/liter	33 (27–38)
Hematuria	153/174 (88)
Proteinuria, median (IQR) gm/day	1.5 (0.6–3)
CRP, median (IQR) mg/liter	27 (8–60)

* Except where indicated otherwise, values are the number/total number (percent) of patients. IQR = interquartile range; eGFR = estimated glomerular filtration rate; CRP = C-reactive protein.

Upon independent review of the renal pathology reports (n = 67), the majority of patients were classified as having either class II glomerulonephritis (33 [49%] of 67 patients; i.e., presence of a focal and segmental glomerulonephritis with segmental endo- and extracapillary proliferation involving <50% of the glomeruli) or class IIIa glomerulonephritis (18 [27%] of 67 patients; i.e., presence of an endocapillary proliferative glomerulonephritis with moderate endocapillary proliferative lesions).

Table 2. Histologic features of the patients with IgA vasculitis*

Skin biopsy	222/260 (85)
Leukocytoclastic vasculitis	205/222 (92)
IgA deposits	174/216 (81)
C3 deposits	47/222 (21)
Fibrinoid necrosis	59/222 (27)
Renal biopsy	144/182 (79)
IgA mesangial deposits	142/144 (99)
Extracapillary proliferation	59/144 (41)
Fibrinoid necrosis	46/144 (32)
Glomerular sclerosis	47/144 (33)
% of glomerular sclerosis, median (IQR)	11 (7–20)
% of interstitial fibrosis, median (IQR)	15 (10–21)
Tubulointerstitial nephritis	44/144 (31)
Glomerulonephritis classification	
Class I	2/67 (3)
Class II	33/67 (49)
Class IIIa	18/67 (27)
Class IIIb	9/67 (13)
Class IV	4/67 (6)
Class V	1/67 (2)

* Except where indicated otherwise, values are the number/total number (percent) of patients. IQR = interquartile range.

Treatments. Treatments used were colchicine in only 27 patients, CS alone in 122 patients, and CS in combination with CYC in 35 patients. Ten patients received various other therapies, including dapsone in 4 patients, rituximab in 2 patients, mycophenolate mofetil in 2 patients, and hydroxychloroquine in 2 patients. Sixty-six patients did not receive a specific treatment.

The baseline characteristics of the 250 patients according to the treatment used (except the 10 patients with various other treatments) are shown in Table 3. Compared to patients who received no treatment or who received only colchicine, patients treated with CS or CS plus CYC were more frequently male ($P = 0.004$). Moreover, patients in the CS or CS plus CYC groups more frequently presented at baseline with necrotic purpura ($P = 0.006$), renal involvement ($P = 0.0001$), higher levels of proteinuria ($P < 0.0001$) and hematuria ($P = 0.0005$), endo- and extracapillary proliferation ($P < 0.0001$), and severe gastrointestinal tract involvement ($P < 0.0001$). These factors, as well as significant factors previously described in the literature, were included in the multivariable logistic regression model and also in the estimation of the propensity score.

Effects of therapeutic regimen on response to treatment and outcome. Of the 260 patients with IgAV, 132 received either colchicine, CS, or CS plus CYC, and had a follow-up time of >6 months. Follow-up data were missing for 5 patients, who did not differ from the remaining patients (data not shown). In total, 127 patients were analyzed for treatment response. The characteristics of the patients according to achievement

Table 3. Characteristics of the patients according to the treatment received*

Characteristic	CS + CYC (n = 35)	CS (n = 122)	Colchicine only (n = 27)	No therapy (n = 66)	P†
Age, mean ± SD years	47 ± 18	51 ± 20	45 ± 16	51 ± 18	0.36
Sex					0.004
Men	31 (89)	76 (62)	15 (56)	35 (53)	
Women	4 (11)	45 (38)	12 (44)	31 (47)	
Skin involvement	35 (100)	122 (100)	27 (100)	66 (100)	
Necrotic purpura	17 (49)	30 (25)	5 (19)	12 (18)	0.006
Fibrinoid necrosis at biopsy	9 (26)	33 (27)	4 (15)	11 (17)	0.28
Joint involvement	23 (66)	79 (65)	16 (59)	32 (48)	0.15
Renal involvement	30 (86)	93 (76)	11 (41)	39 (59)	0.0001
Proteinuria >1 gm/day	24 (69)	53 (43)	4 (15)	17 (26)	<0.0001
Proteinuria >3 gm/day	13 (37)	24 (20)	1 (4)	5 (7)	0.0004
Hematuria	27 (77)	79 (65)	9 (33)	31 (47)	0.0005
Creatinine, median (IQR) μmoles/liter	80 (70–117)	80 (66–111)	75 (66–85)	71 (62–86)	0.02
eGFR, median (IQR) ml/minute/1.73 m^2	90 (50–112)	90 (61–108)	96 (87–105)	98 (80–111)	0.23
Renal biopsy	29 (83)	69 (57)	8 (30)	29 (44)	<0.0001
Endocapillary GN	18 (62)	33 (48)	3 (37)	9 (31)	
Extracapillary GN	14 (48)	35 (51)	0 (0)	5 (17)	
GI involvement	26 (74)	75 (61)	7 (26)	21 (32)	<0.0001
Ileus	3 (9)	8 (7)	0 (0)	2 (3)	0.38
Bleeding	14 (40)	23 (19)	1 (4)	2 (3)	<0.0001
Surgical abdomen	3 (9)	2 (2)	0 (0)	0 (0)	0.054

* Except where indicated otherwise, values are the number (percent) of patients. CS = corticosteroids; CYC = cyclophosphamide; IQR = interquartile range; eGFR = estimated glomerular filtration rate; GN = glomerulonephritis; GI = gastrointestinal.
† P values were for comparison of the CS + CYC and CS groups with the colchicine only and no therapy groups. The P values were determined by chi-square test, except for comparisons of ileus and surgical abdomen involvement, which were by Fisher's exact test.

Table 4. Characteristics of the patients according to achievement of a partial or complete response to therapy in comparison to nonresponders*

Characteristic	Response (n = 97)†	Nonresponse (n = 30)
Age, mean ± SD years	49 ± 19	48 ± 17
Sex		
Men	61 (63)	23 (77)
Women	36 (37)	7 (23)
Skin involvement	97 (100)	30 (100)
Necrotic purpura	30 (31)	9 (30)
Fibrinoid necrosis at biopsy	25 (26)	8 (27)
Joint involvement	65 (67)	16 (53)
Renal involvement	69 (71)	21 (70)
Proteinuria >1 gm/day	46 (47)	16 (53)
Proteinuria >3 gm/day	22 (23)	9 (30)
Hematuria	58 (60)	17 (57)
Creatinine, median (IQR) μmoles/liter	76 (67–104)	88 (78–110)
eGFR, median (IQR) ml/minute/1.73 m^2	95 (65–111)	86 (57–96)
Renal biopsy	59 (61)	19 (63)
Endocapillary GN	31 (53)	11 (58)
Extracapillary GN	30 (51)	8 (42)
GI involvement	55 (57)	15 (50)
Ileus	4 (4)	1 (3)
Bleeding	28 (29)	4 (13)
Surgical abdomen	2 (2)	1 (3)
First-line treatment		
CS (n = 80)	64 (66)	16 (53)
CS + CYC (n = 30)	23 (24)	7 (23)
Colchicine (n = 17)	10 (10)	7 (23)

* Except where indicated otherwise, values are the number (percent) of patients. IQR = interquartile range; eGFR = estimated glomerular filtration rate; GN = glomerulonephritis; GI = gastrointestinal; CS = corticosteroids; CYC = cyclophosphamide.
† None of the *P* values versus the nonresponse group were significant.

of a partial or complete response to therapy within the first 12 months of follow-up, in comparison to those with no response, are summarized in Table 4.

In univariate analysis, no characteristic or treatment was significantly associated with the achievement of a partial or complete response to treatment. A partial or complete response was observed in 64 (80%) of 80 patients treated with CS, 23 (77%) of 30 patients treated with CS plus CYC, and 10 (59%) of 17 patients treated with colchicine ($P = 0.17$).

Among the 66 patients who did not receive any treatment, 34 were available for further analysis because they were followed up for at least 6 months. In univariate analysis, 29 (85%) of the 34 patients who did not receive treatment experienced spontaneous disease remission at months 6 and/or 12. Although patients were not comparable in each treatment group, the cutaneous manifestations persisted in 7–41% of patients between month 6 and month 12 of follow-up, and the renal manifestations persisted in 10–40% of patients between month 6 and month 12 among the various treatment groups (see Supplementary Table 2, available on the *Arthritis & Rheumatology* web site at http://onlinelibrary.wiley.com/doi/10.1002/art.40178/abstract).

We then evaluated the efficacy of treatments after adjusting for confounding factors in a multivariable logistic regression model, with adjustments for sex, creatinine level, proteinuria level, presence of digestive bleeding, and presence of necrotic purpura. Results of the multivariable propensity score–adjusted logistic regression model and the inverse probability weighting on propensity score model, with or without sensitivity analysis excluding outliers, are shown in Figures 1 and 2.

Treatment with CS or CS plus CYC was more effective than colchicine in achieving a partial or complete response. This was shown using the multivariable logistic regression model (OR 3.68, 95% CI 1.10–12.33; $P = 0.03$), the multivariable propensity score–adjusted logistic regression model (OR 3.58, 95% CI 1.07–11.94; $P = 0.04$), and the inverse probability weighting on propensity score model (OR 3.75, 95% CI 1.28–10.99; $P = 0.02$) (Figure 1).

We also compared the efficacy of CS plus CYC to that of CS alone and found that the results were discordant according to the different statistical methods used (Figure 2). Use of the multivariable logistic regression model and the multivariable propensity score–adjusted logistic regression model did not demonstrate any difference between the 2 therapeutic strategies in terms of achieving a partial or complete response (OR 0.88, 95% CI 0.29–2.67 [$P = 0.82$] for the CS alone group versus OR 0.90, 95% CI 0.29–2.78 [$P = 0.86$] for those receiving CS plus CYC). In contrast, use of inverse probability weighting on propensity score, with or without sensitivity analysis excluding outliers, showed that treatment with CS plus CYC was more effective than treatment with CS alone in achieving a partial or complete response (with sensitivity analysis, OR 1.79, 95% CI 1.00–3.20 [$P = 0.049$]; without sensitivity analysis, OR 2.33, 95% CI 1.29–4.18 [$P = 0.005$]).

Follow-up of patients. After a median follow-up of 17.2 months (range 9.1–38.3 months), corresponding to 593 patient-years, 8 patients died, including 3 deaths directly related to IgAV (2 from mesenteric ischemia and 1 from multivisceral failure). Eight patients experienced end-stage renal failure that was treated with renal transplantation (n = 2) or dialysis (n = 6). Among patients who received any of the treatments, data concerning relapse during the first 12 months after treatment were available for 107 patients. These data showed that 15 minor relapses (14% of patients) and 9 major relapses (8% of patients) occurred. Among the untreated patients, data concerning relapse during the first 12 months of follow-up were available for 10 patients. In this group, 1 minor relapse (10% of patients) occurred.

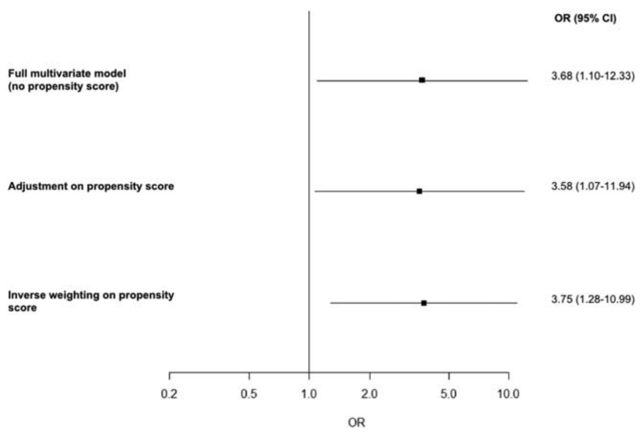

Figure 1. Efficacy of corticosteroids or corticosteroids plus cyclophosphamide relative to colchicine for the treatment of adult patients with IgA vasculitis (Henoch-Schönlein) as assessed using a full multivariable logistic regression model (with no propensity score), a multivariable propensity score–adjusted logistic regression model, and an inverse probability weighting on propensity score model. Squares with horizontal lines show the odds ratio (OR) with 95% confidence interval (95% CI).

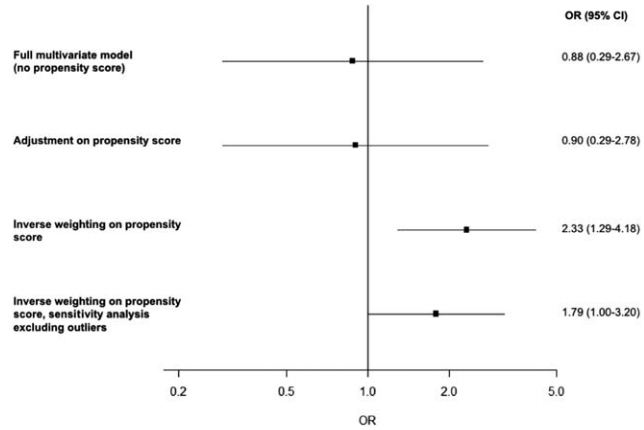

Figure 2. Efficacy of corticosteroids plus cyclophosphamide relative to corticosteroids alone for the treatment of adult patients with IgA vasculitis (Henoch-Schönlein) as assessed using a full multivariable logistic regression model (with no propensity score), a multivariable propensity score–adjusted logistic regression model, an inverse probability weighting on propensity score model, and an inverse probability weighting on propensity score model with a sensitivity analysis in which outliers were excluded. Squares with horizontal lines show the odds ratio (OR) with 95% confidence interval (95% CI).

DISCUSSION

It has been clearly reported that the clinical presentation and prognosis of adults with IgAV differs from that of children. Because randomized controlled trials are lacking in the literature, evaluation of the efficacy of the different therapeutic regimens, which is mandatory to improve the management of IgAV, was a main goal of the French multicenter and transdisciplinary IGAVAS survey. To better define the clinical spectrum and the therapeutic management of IgAV, we analyzed the data from the 260 patients included in the IGAVAS survey, which constitutes the largest series of adults with IgAV reported in the literature so far.

Patients in our study were comparable to those previously described by other investigators (16–20). The study patients were predominantly male, and the mean age was 50 years, as compared to ages 32–44 years in previous series (16–20) and a mean age of 48 years in a northwestern Spanish epidemiologic study (5). The incidence of cutaneous, articular, and gastrointestinal tract involvement was also comparable to that described in other series, whereas renal involvement was more prevalent in our series (70% versus 30–60% in other series) (16–20). This finding is probably related to the high percentage of patients included by nephrologists in our survey. In considering the laboratory findings, the prevalence of increased serum IgA levels (53% of patients) was slightly more notable than that in other series (prevalence of 31–52%) (16–20).

Treatment of IgAV is often based on symptoms, because the disease course is usually benign. In more severe forms of the disease, some studies have evaluated the use of CS and immunosuppressive agents (21), but randomized controlled trials are lacking in adult patients with IgAV. CS agents are effective on arthralgias and abdominal pain, but there is considerable controversy with regard to the benefit of CS to treat renal involvement and prevent evolution to ESRD, both in pediatric and adult populations (22–25). Findings from a study by Ren et al in adult patients with IgAV and severe nephritis suggested that mycophenolate mofetil could be useful for inducing remission and as a steroid-sparing agent, but the long-term effects on renal function are not known (26).

By analogy, CYC has been used to treat patients with severe autoimmune diseases, particularly those with organ- or life-threatening manifestations of IgAV. Pillebout et al compared treatment with CS alone to treatment with CS in combination with CYC in adults with severe IgAV in a prospective open-label trial (11). Fifty-four patients with biopsy-proven IgAV and severe manifestations were included and randomized to receive CS alone or CS plus CYC. At 6 months, no difference was found between the 2 groups with regard to the primary end point, i.e., the achievement of complete disease remission (defined as a score of 0 on the Birmingham Vasculitis Activity Score) (27), with no persistent or new clinical and/or biologic manifestations of vasculitis. The secondary end points, i.e., renal outcome, deaths, and adverse events, also did not differ at 12 months. However, only 54 patients of the 200 initially planned were included, and thus the data should be interpreted with caution. In addition, overall survival at 12 months tended to be better in the CYC combination treatment group, with an overall survival of 96% in the CS plus CYC treatment arm compared to 79% with CS alone ($P = 0.08$). Therefore, the use of CYC as a therapy for IgAV remains controversial. Recommendations from the Kidney Disease Improving Global Outcomes Group suggest that IgAV-related nephritis in adults should be treated in a manner similar to that in children, and that CYC should not be used except to treat crescentic glomerulonephritis (crescents in >50% of the glomeruli) with nephrotic syndrome or rapid degradation of the eGFR (28).

The descriptive analysis of real-life patients may also provide interesting findings on the efficacy of the different therapeutic regimens. In this study, we used several complementary approaches to compare treatment efficacy. First, we did a multivariable analysis taking into account potential confounding factors, which was not frequently done in previous studies because of the limited sample sizes of those studies. We also used a propensity score to take into account a potential indication bias, as patients treated with CS plus CYC may have more severe disease at baseline than those treated with CS alone or with colchicine (15). We took the propensity score into account in analyses with both adjustment and ponderation by inverse variance. All approaches showed concordant results for the comparison between CS or CS plus CYC and colchicine. Furthermore, our cohort may be biased toward more severe disease, and the prevalence and course of less serious disease cannot be interpreted to the same extent as more serious disease.

However, this was not the case for the comparison between CS alone and CS plus CYC, since we obtained contradictory results according to the statistical methods used. Indeed, the multivariable logistic regression model and the multivariable propensity score–adjusted logistic regression model did not demonstrate any difference between the 2 therapeutic strategies. In contrast, the use of inverse probability weighting on propensity score with or without sensitivity analysis showed that CS plus CYC was more effective than CS alone in achieving a partial or complete response. Propensity score weighting is a frequently recommended approach, but could be sensitive to outliers, and therefore no definitive conclusion can be taken for this comparison (13,14). In addition, we cannot exclude the possibility of bias related to unknown confounders that were not taken into account in the propensity score.

These conflicting results indicate that, besides the limitations related to the retrospective design of this study and possibly the insufficient power to detect any difference, no firm conclusion can be drawn regarding the comparison between CS alone and CS plus CYC. In addition, the low number of patients receiving colchicine alone or CS plus CYC, as well as the number of nonresponder patients, may have limited the power of the study. However, given the potential for adverse events related to the adjunction of CYC in this condition, treatment with CS alone seems to be a reasonable option for first-line therapy in patients with systemic IgAV, except in very severe presentations, for which decisions should be made individually.

In conclusion, this series constitutes the largest series of adults with IgAV reported in the literature so far. It provides interesting data on the clinical and histologic presentation of the patients and therapeutic efficacy, showing that CS alone appears to be a reasonable first-line therapy in patients with systemic IgAV, while the use of CYC remains controversial.

ACKNOWLEDGMENTS

The authors thank Corinne Alberti (Paris), Emmanuel Andrès (Strasbourg), Jean-Benoit Arlet (Paris), Denis Bagneres (Marseille), Sabine Berthier (Dijon), Benoît Brihaye (St. Quentin), Philippe Evon (Bar-le-duc), Sophie Georgin-Lavalle (Paris), Nicolas Girszyn (Rouen), Bruno Gombert (La Rochelle), Maxence Ficheux (Caen), Thierry Lobbedez (Caen), Véronique Le Guern (Paris), Jean Christophe Lega (Lyon), Serge Madaule (Albi), Rosine Mangouka (Chartres), Camille Martinez (Strasbourg), Nadine Meaux-Ruault (Metz), Luc Mouthon (Paris), Laurent Perard (Lyon), Guillaume Queffeulou (Cherbourg), Alexis Regent (Paris), Françoise Sarrot-Reynauld (Grenoble), Laurent Chiche (Marseille), Jérome Février (Caen), Pierre-Yves Hatron (Lille), Eric Hachulla (Lille), Marc Lambert (Lille), Cécile Morice (Caen), Nicolas Bouvier (Caen), and Laurence Verneuil (Caen).

AUTHOR CONTRIBUTIONS

All authors were involved in drafting the article or revising it critically for important intellectual content, and all authors approved the final version to be published. Dr. Terrier had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

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